

AP CALCULUS AB – UNIT 3

Differentiation: Composite, Implicit, and Inverse Functions

AP-Style Review and Exam Practice



Name: _____ Class: _____ Date: _____

Directions

Section I Part A contains 16 non-calculator multiple-choice questions. Section I Part B contains 8 calculator-permitted multiple-choice questions. Section II contains four free-response questions. Show setup, exact reasoning, units, and justification.

Section I – Part A: No Calculator

1. Differentiate $f(x) = (3x - 1)^3$.

(A) $9x - 3$

(B) $3(3x - 1)^2$

(C) $9(3x - 1)^2$

(D) $9(3x - 1)^3$

2. Find $\frac{d}{dx}[\sin(5x^2)]$.

(A) $10x \cos(5x^2)$

(B) $5 \cos(5x^2)$

(C) $\cos(5x^2)$

(D) $10x \sin(5x^2)$

3. Differentiate e^{5x^3} .

(A) $15x^2$

(B) $15x^2 e^{5x^3}$

(C) $15x^2 e^{5x}$

(D) e^{5x^3}

4. Differentiate $\ln(x^2 + 1)$.

(A) $\frac{2}{x^2+1}$

(B) $2x \log(x^2 + 1)$

(C) $\frac{1}{x^2+1}$

(D) $\frac{2x}{x^2+1}$

5. For the implicit curve $x^2 + xy + y^2 = K$, which formula for dy/dx is correct?

(A) $-\frac{2x+y}{x+2y}$

(B) $-\frac{x+2y}{2x+y}$

(C) $-\frac{2x+1}{1+2y}$

(D) $\frac{2x+y}{x+2y}$

6. For $x^2 + y^2 = 13$, find the slope at $(2, 3)$.

(A) $\frac{3}{2}$

(B) $-\frac{3}{2}$

(C) $-\frac{2}{3}$

(D) $\frac{2}{3}$

7. If $f(1) = 4$ and $f'(1) = 2$, find $(f^{-1})'(4)$.

(A) $-\frac{1}{2}$

(B) $\frac{1}{2}$

(C) 2

(D) $\frac{1}{3}$

8. Differentiate $\arcsin(4x)$.

- (A) $\frac{4}{1+16x^2}$ (B) $\frac{4}{\sqrt{1-16x^2}}$
(C) $\frac{1}{\sqrt{1-16x^2}}$ (D) $-\frac{4}{\sqrt{1-16x^2}}$

9. Differentiate $\arctan(x^2)$.

- (A) $\frac{2x^1}{1+x^2}$ (B) $\frac{2x}{x^4+1}$
(C) $\frac{1}{1+x^4}$ (D) $\frac{2x^1}{\sqrt{1-x^4}}$

10. Differentiate $x(x^2 + 2)^4$.

- (A) $(x^2 + 2)^4$ (B) $4x(x^2 + 2)^3$
(C) $(x^2 + 2)^3(9x^2 + 2)$ (D) $8x(x^2 + 2)^3$

11. Differentiate $f(x) = (4x - 1)^3$.

- (A) $3(4x - 1)^2$ (B) $12(4x - 1)^3$
(C) $16x - 4$ (D) $12(4x - 1)^2$

12. Find $\frac{d}{dx}[\sin(3x^2)]$.

- (A) $3 \cos(3x^2)$ (B) $6x \cos(3x^2)$
(C) $\cos(3x^2)$ (D) $6x \sin(3x^2)$

13. Differentiate e^{4x^3} .

- (A) $12x^2$ (B) $12x^2 e^{4x}$
(C) $12x^2 e^{4x^3}$ (D) e^{4x^3}

14. Differentiate $\ln(x^2 + 3)$.

- (A) $\frac{1}{x^2+3}$ (B) $2x \log(x^2 + 3)$
(C) $\frac{2x}{x^2+3}$ (D) $\frac{2}{x^2+3}$

15. For the implicit curve $x^2 + xy + y^2 = K$, which formula for dy/dx is correct?

- (A) $-\frac{2x+1}{1+2y}$ (B) $\frac{2x+y}{x+2y}$
(C) $-\frac{x+2y}{2x+y}$ (D) $-\frac{2x+y}{x+2y}$

16. For $x^2 + y^2 = 29$, find the slope at $(2, 5)$.

- (A) $\frac{5}{2}$ (B) $-\frac{2}{5}$
(C) $-\frac{5}{2}$ (D) $\frac{2}{5}$

Section I – Part B: Calculator Permitted

17. Use a calculator to evaluate the derivative of $\sin(2x)$ at $x = -0.66$.
- (A) 0.248 (B) 0.496
(C) -0.496 (D) -1.937
18. Find $\frac{d}{dx}(\arcsin x)$ at $x = -0.25$ to three decimals.
- (A) 0.75 (B) -1.033
(C) -0.253 (D) 1.033
19. A numerical table gives $f(3.0) = 4.6$ and $f'(3.0) = 2.0$. Estimate $(f^{-1})'(4.6)$.
- (A) 0.5 (B) 0.217
(C) -0.5 (D) 2.0
20. For $x^2 + xy + y^2 = K$, estimate dy/dx at $(1.94, 1.55)$.
- (A) -0.928 (B) -1.252
(C) 1.077 (D) -1.077
21. Use a calculator to evaluate the derivative of $\sin(4x)$ at $x = -1.04$.
- (A) -0.525 (B) -2.099
(C) 2.099 (D) 3.405
22. Find $\frac{d}{dx}(\arcsin x)$ at $x = 0.56$ to three decimals.
- (A) 0.594 (B) 1.207
(C) -1.207 (D) 1.56
23. A numerical table gives $f(2.9) = 6.8$ and $f'(2.9) = 0.8$. Estimate $(f^{-1})'(6.8)$.
- (A) -1.25 (B) 1.25
(C) 0.147 (D) 0.8
24. For $x^2 + xy + y^2 = K$, estimate dy/dx at $(1.01, 1.11)$.
- (A) -0.969 (B) -1.032
(C) -0.91 (D) 0.969

Section II – Free Response

Questions 1–2: Calculator permitted. Questions 3–4: No calculator.

FRQ1 – Nested Chain Rule

Let $f(x) = e^{(x^2+1)^3}$.

(a) Identify the three nested layers. [2 points]

(b) Find $f'(x)$. [3 points]

(c) Evaluate $f'(0)$. [1 points]

FRQ2 – Implicit curve

The curve $x^2 + xy + y^2 = 7$ contains the point $(1, 2)$.

(a) Find dy/dx . [3 points]

(b) Find the tangent slope at $(1, 2)$. [1 points]

(c) Write the normal line at the point. [2 points]

FRQ3 – Derivative of an inverse

A one-to-one differentiable function satisfies $f(2) = 5$ and $f'(2) = -3$.

(a) Find $f^{-1}(5)$. [1 points]

(b) Find $(f^{-1})'(5)$. [2 points]

(c) Explain the geometric meaning of the reciprocal slope. [2 points]

FRQ4 – Inverse trigonometric composition

Let $g(x) = \arctan(1 + x^2)$.

(a) Find $g'(x)$. [3 points]

(b) Determine where g is increasing. [2 points]

(c) Find $g'(1)$. [1 points]